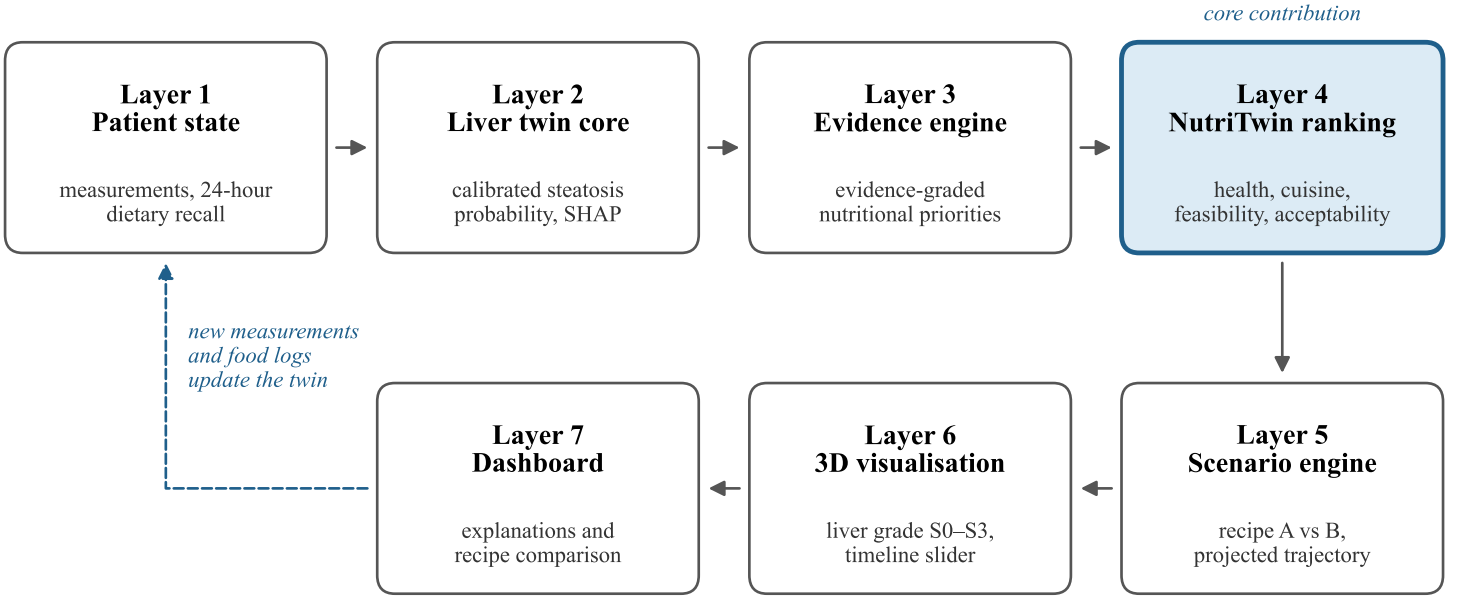
core contribution
Layer 1
Patient state
measurements, 24-hour dietary recall
Layer 2
Liver twin core
calibrated steatosis probability, SHAP
Layer 3
Evidence engine
evidence-graded nutritional priorities
Layer 4
NutriTwin ranking
health, cuisine, feasibility, acceptability
Layer 5
Scenario engine
recipe A vs B, projected trajectory
Layer 6
3D visualisation
liver grade S0–S3, timeline slider
Layer 7
Dashboard
explanations and recipe comparison
new measurements and food logs update the twin